PATHOGENESIS/PATHOPHYSIOLOGY (normal → disease state)

1. Explain how insulin, glucagon and catecholamines regulate the metabolism of carbohydrates, lipids, and protein. BCM - **Spencer**
2. Compare and contrast the pathophysiological mechanisms of DKA and HHS. CM **Connor**
3. JP: Outline the current understanding of the pathophysiological relationships between obesity, dyslipidemia, hyperglycemia and hypertension. CM DONE
4. Compare and contrast the pathophysiology of Type 2 diabetes, Type 1 diabetes, MODY, and diseases of the exocrine pancreas (e.g., pancreatitis, cystic fibrosis). CM **Ravi**
5. Distinguish autoimmune mechanisms of Type 1 diabetes from inflammatory mechanisms associated with Type 2 diabetes. INIM **Connor**
6. Differentiate the histologic features seen in the pancreas in type 1 diabetes versus type 2 diabetes. PATH- **Tomiwa**
7. JP: Describe the pathologic changes of chronic diabetes complications as manifested in the vessels, kidneys, eyes, and nerves. PATH DONE
8. JP: Explain how hyperglycemia affects the immune response and contributes to increased susceptibility to infections. INIM DONE
9. Explain the effect of acute hyperglycemia on cognition. BCM **Shannon (Done)**

EPIDEMIOLOGY (incidence, prevalence, risk factors)

1. Identify the risk factors for type 2 diabetes and summarize the recommendations regarding screening among asymptomatic individuals. CM and EBM **Sophie** Done
2. Describe the epidemiology of diabetes in the United States. CM **Sophie Done**

CLINICAL PRESENTATION (signs, symptoms, history, exam)

1. Describe and interpret the common complications of diabetes seen on physical exam (ex: funduscopic exam, peripheral neuropathies, skin exam findings). CS **Ishaan done**
2. Compare and contrast the clinical presentation for type 1 and type 2 diabetes. CM- **Tomiwa**
3. Compare and contrast the clinical presentations of the diabetic emergencies (DKA vs HHS). CM - **Ayla P**

DIAGNOSIS (differential diagnosis, diagnostic tests and tools, clinical reasoning)

1. Recognize values for plasma glucose, bicarbonate, ketone bodies, osmolality, and blood gases that are typical of HHS. BCM - **Spencer**
2. List and recognize the current ADA criteria for the diagnosis of diabetes. CM **Shannon (done)**
3. Define the condition of prediabetes and describe the risk of progression to diabetes. CM **Sophie Done**

TREATMENT AND OUTCOMES

1. Describe the major effect of metformin on carbohydrate metabolism. BM **Shannon (Done)**
2. Compare and contrast the mechanisms of action, relative efficacies, adverse effects, and contraindications for drug classes used to treat type 2 diabetes. PHARM **Ishaan done**
3. Explain how diabetes prevention and management contribute to healthcare value. HSS (attos://www.cdc.gov/nccdphp/priorities/diabetes- interventions.htmi#fedc generic section 3-strategies-that-work) - **Ayla P**
4. Describe lifestyle strategies and their benefits in the diabetic and prediabetic patient. WELL (see additional resources) - **Ayla P**
5. Describe the major acute and chronic diabetic complications that are the result of poor glucose control. CM **Ravi**

Explain how insulin, glucagon and catecholamines regulate the metabolism of carbohydrates, lipids, and protein. BCM - Spencer

1. Insulin: Anabolic Storage Hormone

Source/trigger: pancreatic β-cells, released when blood glucose rises (fed state)
Targets: liver, skeletal muscle, adipose
Signaling: insulin receptor (tyrosine kinase) → IRS1/2 → PI3K → Akt

- GSK3 inhibition → glycogen synthesis
- FOXO inhibition → ↓ gluconeogenesis
- GLUT4 translocation → glucose uptake (muscle, fat)
- mTORC1 → protein synthesis

Carbohydrate (↓ blood glucose)

- ↑ GLUT4-mediated glucose uptake in muscle and adipose
- ↑ Glycolysis (↑ hexokinase, ↑ PFK)
- ↑ Glycogen synthesis (↑ glycogen synthase)
- ↓ Hepatic glycogenolysis and ↓ gluconeogenesis

Lipid (storage)

- ↓ Lipolysis by inhibiting hormone-sensitive lipase (HSL) → ↓ plasma FFA
- ↑ Fatty acid and triglyceride synthesis
- ↑ Lipoprotein lipase → FA uptake into adipocytes

Protein (anabolic)

- ↑ Amino acid uptake
- ↑ Protein synthesis (mTORC1)
- ↓ Proteolysis

2. Glucagon: Hepatic Mobilizing Hormone

Source/trigger: pancreatic α-cells, released in hypoglycemia/fasting

- Also stimulated by amino acids (protein meal)
  Target: almost exclusively liver
- No meaningful receptors on muscle or fat, so no effect on muscle glucose uptake
  Signaling: glucagon receptor (GCGR; Gαs/Gq) → ↑ cAMP → PKA, plus ↑ Ca^{2+}

Carbohydrate (↑ blood glucose, the primary action)

- ↑ Hepatic glycogenolysis: rapid, short-term, cAMP-mediated
- ↑ Gluconeogenesis: sustained, via ↑ PEPCK and other rate-limiting enzymes

Lipid (hepatic)

- ↑ β-oxidation (↑ CPT-1 / carnitine acyltransferase-1)

- ↓ Lipogenesis
    - Inhibits acetyl-CoA carboxylase → ↓ malonyl-CoA, which relieves CPT-1 inhibition
- ↑ PPARα
- A direct effect on adipose lipolysis in humans is controversial, since GCGR is not clearly expressed in human fat.

Protein

- ↑ Hepatic amino acid uptake and catabolism
    - Carbon skeletons → gluconeogenesis
    - Amino groups → urea cycle
- Gluconeogenesis and ureagenesis are coupled.

Liver–α-cell axis (feedback loops)

1. Glucose loop: ↓ glucose → ↑ glucagon → ↑ hepatic glucose output, and vice versa.
2. Amino acid loop: ↑ plasma AAs → ↑ glucagon → ↑ hepatic AA uptake and ureagenesis.
3. Possible lipid loop: glucagon ↑ FA oxidation. Whether plasma FFA regulates glucagon secretion is unclear.

3. Catecholamines: Stress/Sympathetic Hormones

Source: epinephrine from the adrenal medulla; norepinephrine from sympathetic nerves
Targets: liver, muscle, and adipose (broader than glucagon)
Signaling:

- β-receptors → Gs → ↑ cAMP → PKA (same pathway as glucagon)
- α1 → Gq → PLC → IP_3 → ↑ Ca^{2+}
  Context: acute stress (“fight or flight”) and exercise

Carbohydrate (↑ blood glucose)

- ↑ Hepatic glycogenolysis and gluconeogenesis
- ↑ Skeletal muscle glycogenolysis and glycolytic flux
    - Muscle glucose is used locally because muscle lacks glucose-6-phosphatase.
- ↓ Insulin secretion (via α2 on β-cells)
- ↓ Insulin-stimulated glucose uptake, sparing glucose for the brain and working muscle

Lipid (mobilization)

- Potent ↑ adipose lipolysis (β-adrenergic, including β3) → activates HSL → ↑ plasma FFA for fuel
- ↑ Thermogenesis
- Oppose insulin’s antilipolytic action

Protein

- Generally catabolic as part of substrate mobilization
- Less prominent than their effects on glucose and lipid

Compare and contrast the pathophysiological mechanisms of DKA and HHS. CM

Diabetic Ketoacidosis (DKA)

- More common in T1DM

Etiology:

- absence of insulin
- insulin treatment failing for known diabetes
- infection (most common)
- other illness (pancreatitis, MI)
- surgery or trauma
- drugs/alcohol use

Pathophysiology:

- Acidosis: no insulin→ glucagon → increase in glucose → lipolysis → FAs go to the liver → become ketone bodies (acetoacetic acid and beta-hydroxybutyric acid) → acidosis because of the ketone body accumulation in blood.
    - **Kussmaul's respirations** to reduce CO2 and compensate for acidity **(heavy, fast breathing)**
    - hyperkalemia in serum because
        - blood acidic so increase in H+ ions → influx of H+ through H+/K+ exchanger
        - insulin normally drives K into the cell
    - high anion gap because of the all the ketone bodies dissociated into H+ and anion → ↓ **pH**

- Polyuria: because inability of kidneys to filter high glucose + ketone body levels → excretion of glucose + ketone bodies into urine → hyperosmolarity → water + electrolytes (Na^+, K^+) shift out of cells and into urine causing hypovolemia.

- High stress states: release epinephrine → release glucagon → elevates blood sugars level → loss of glucose in urine again → loss of water → dehydration → need for alternative energy → ketone bodies made → ketoacidosis.
        - Ketones break down into acetone in lungs → sweet fruity breath
        - Ketoacidosis → N/V, mental status changes, cerebral edema

- Hyperglycemia:
    - **Maximal glucose production:** no inhibition of glycogenolysis or gluconeogenesis
    - **Minimal glucose use:** no stimulation of glycolysis, glycogenesis, or GLUT-4; replacement of glucose w/ fatty acids as an energy source

Hyperosmotic Hyperketotic Syndrome (HHS)

- more common T2DM

Pathophysiology:

- Hyperglycemia:
    - High glucose production: no inhibition of glycogenolysis or gluconeogenesis by insulin; **stimulation of gluconeogenesis by counterregulatory hormones***
    - Low glucose use: no stimulation of glycolysis, glycogenesis, or GLUT-4 by insulin; replacement of glucose w/ fatty acids as an energy source

- Dehydration: hyperglycemia → inability of kidneys to filter high glucose levels → excretion of glucose into urine → water follows glucose (hygroscopic)

Obesity is the primary upstream driver linking dyslipidemia, hyperglycemia, and hypertension, mainly through visceral adipose tissue dysfunction and systemic insulin resistance.

- Excess visceral adiposity impairs insulin's antilipolytic effects, increasing lipolysis and circulating free fatty acids (FFAs) which accumulate in the liver and skeletal muscle, disrupting insulin receptor signaling and causing insulin resistance
- Pancreatic β-cells initially compensate by ↑ insulin secretion (hyperinsulinemia). Over time, metabolic stress and lipotoxicity cause progressive β-cell failure, so insulin secretion can no longer compensate for insulin resistance (hepatic gluconeogenesis remains unsuppressed and muscle glucose uptake ↓) producing **hyperglycemia and Type 2 diabetes**
- ↑ FFA delivery to the liver combined with hepatic insulin resistance increases VLDL, LDL and triglyceride production while decreasing High-Density Lipoprotein (HDL) cholesterol **= characteristic dyslipidemia triad**
- Compensatory hyperinsulinemia ↑ renal tubular sodium reabsorption and activates the sympathetic nervous system and RAAS while elevated FFAs and hyperglycemia deplete nitric oxide, causing endothelial dysfunction and impaired vasodilation which leads to **systemic vascular resistance and arterial stiffness (hypertension!)**

These conditions reinforce one another and form a self-perpetuating cycle that drastically increases the risk of cardiovascular events, heart failure, and chronic kidney disease.

1. Compare and contrast the pathophysiology of Type 2 diabetes, Type 1 diabetes, MODY, and diseases of the exocrine pancreas (e.g., pancreatitis, cystic fibrosis).

	T1DM	T2DM	MODY	Acute pancreatitis	Chronic pancreatitis	CF (CFRD)
Defect	Autoimmune β-cell destruction	Insulin resistance + β-cell failure	Single-gene β-cell secretion defect	Inflammation / necrosis	Fibrosis of acini + islets	CFTR → thick secretions → d blockage
Mechanism	CD8+ T cells → insulitis	Variable ; Ectopic fat (DAG) → PKC → IRS-1 serine-P → ↓ GLUT4; β-cell exhaustion (gluco/lipotoxicity, amylin)	GCK: ↑ glucose set point; HNF: ↓ β-cell gene transcription	Stress hormones ↑ glucose; necrosis → islet loss	Stellate cells → fibrosis → ↓ insulin, glucagon, PP	Fibrosis/fatty replacement + intrinsic β-cell dysfunction
Genetics	HLA DR3/DR4	Polygenic	AD: GCK, HNF1A, HNF4A, HNF1B	Gallstones, EtOH, ↑ TG	EtOH; PRSS1, SPINK1	AR CFTR (ΔF50
Insulin / C-pep	↓↓	↑ early → ↓ late	Normal	Transient ↓	↓	Delayed, ↓ 1st phase
Glucagon	↑ (relative)	↑	Normal	↑	↓	↓
Insulin resistance	–	+++	–	Transient	Hepatic (↓ PP)	With infection steroids
Acute comp.	DKA	HHS	Rare	Stress hyperglycemia	Hypoglycemia; DKA rare	DKA rare
Classic pt	Young, thin, rapid onset	Older, obese	<25, thin, 3-gen family hx	Epigastric pain → back	EtOH, steatorrhea, chronic pain	Wt loss, ↓ PFTs
Dx	Antibodies: GAD65, IA-2, insulin, ZnT8; ↓ C-pep	Antibody neg, normal/↑ C-pep	Ab neg, C-pep +, genetic testing	Lipase >3x ULN	CT calcifications, ↓ fecal elastase	Yearly OGTT fr age 10 (not HbA1c)
Key points	No glucagon rise in hypoglycemia	1st phase lost first; some insulin left → no ketosis	GCK: mild stable FPG. HNF1A: glycosuria, SU-sensitive. HNF4A: macrosomia. HNF1B: renal cysts	Can → permanent DM after necrosis	Brittle DM; ↓ incretins; often mislabeled T2	Postprandial ↑ glucose first
Tx	Insulin	Metformin, GLP-1 RA, SGLT2i → insulin	GCK: none; HNF1A/4A: SU	Supportive fluids ± insulin	Insulin + enzymes	Insulin

Distinguish autoimmune mechanisms of Type 1 diabetes from inflammatory mechanisms associated with Type 2 diabetes. INIM

Type 1 DM: insulin dependent

- Autoimmune disorder leading to production of autoantibodies that destroy beta cells and thus cause insulin deficiency
- Type IV HS rxn = **T-cell mediated**
- Autoantibody markers: GAD65, IA2, zinc transporter 8 antibody
- Often set off by an environmental trigger in an already genetically susceptible person
    - **Associated with HLA DR3 and HLA DR4**
- With long-standing T1DM, lose counterregulatory hormones → insulin-induced hypoglycemia
    - <1 mo: counterregulation is normal
    - 1-5 yr: glucagon production is decreased while epinephrine production is sustained
    - 14-31 yr: glucagon production is almost gone and epinephrine production is suppressed

Type 2 DM: insulin resistant

- *Mechanism 1: peripheral insulin resistance*
    - Obesity → increased fatty acids → impaired insulin-dependent glucose uptake
        - Need more insulin to achieve adequate glucose removal
    - Mechanism for decreased glucose uptake:
        - NORMALLY, insulin receptor substrate (IRS) activates PI3K pathway → increases GLUT4 expression → increases glucose uptake
        - Increased fatty acids → increased serine kinase activity in fat and skeletal muscle → phosphorylation of IRS → decreased affinity of IRS for PI3K → decreased expression of GLUT4 transporters → decreased glucose uptake
- *Mechanism 2: beta-cell dysfunction*
    - Pro-amylin builds up in the pancreas → decreases endogenous insulin production

	TYPE 1	TYPE 2
Insulin Resistance	No	Yes
Islet Cell Antibodies & GAD-65 Antibodies	Yes	No
Insulin Secretion Defect	Yes	Yes
HLA-Associated Antigens	Yes	No
Associated with other Auto-immune Diseases	Yes	No

1. Differentiate the histologic features seen in the pancreas in type 1 diabetes versus type 2 diabetes. PATH- **Tomiwa**

Feature	T1DM	T2DM
Main lesion	β-cell **destruction**	β-cell **dysfunction** + islet amyloid deposition
Islets	Reduced/atrophic β-cell population	Islets may contain amyloid (IAPP/amylin)
Inflammation	Lymphocytic infiltration (insulitis) can be seen during active autoimmune destruction	Not characterized by autoimmune insulitis
β-cells	Progressively destroyed → severe loss	Initially increased insulin production, later β-cell failure
Amyloid	Not the characteristic lesion	Islet amyloid deposition is characteristic
Result	Absolute insulin deficiency	Relative insulin deficiency due to insulin resistance + β-cell failure

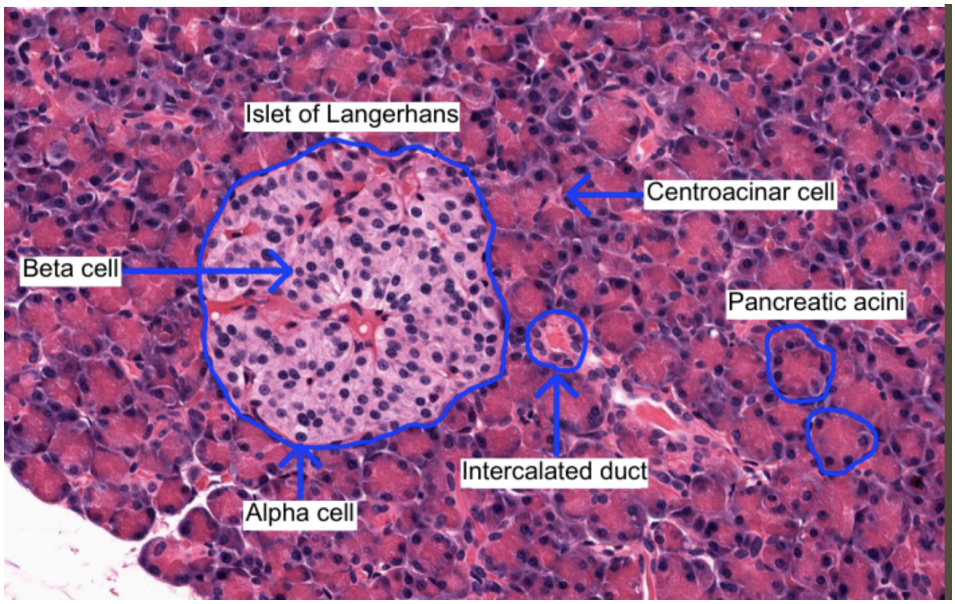
Type 1 DM: think "destroyed β-cells"

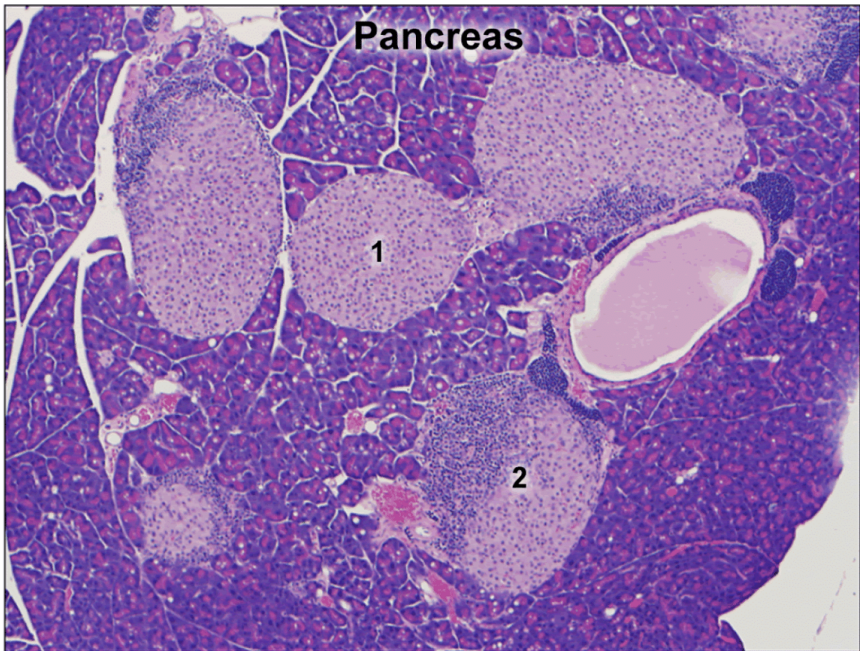
Type 1 DM: Autoimmune attack → lymphocytic inflammation → progressive β-cell destruction → ↓ β-cell mass → absolute insulin deficiency

Type 2 DM: think "amyloid + exhausted β-cells"

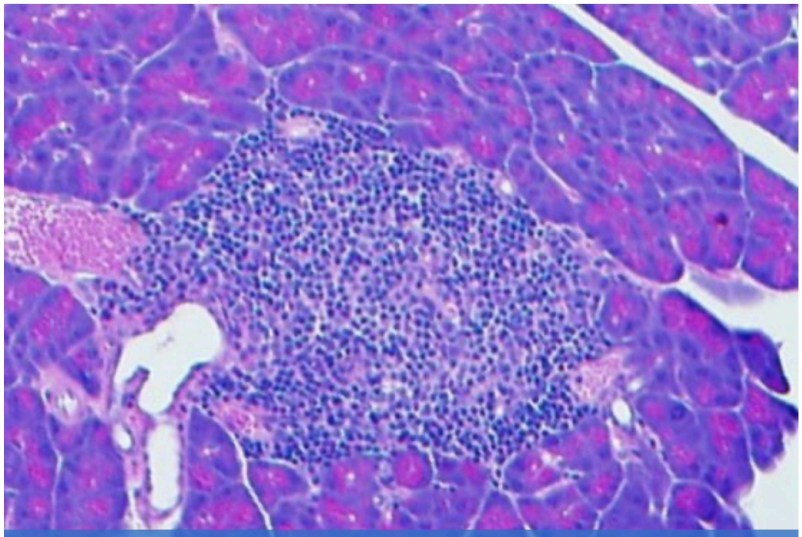
In T2DM, chronic insulin resistance causes the β-cells to compensate by producing more insulin.

Over time, insulin resistance → ↑ demand on β-cells → increased insulin/pro-amylin production → IAPP (amylin) accumulation → islet amyloid → β-cell dysfunction/failure


Islet of Langerhans
Centroacinar cell
Beta cell
Pancreatic acini
Intercalated duct
Alpha cell

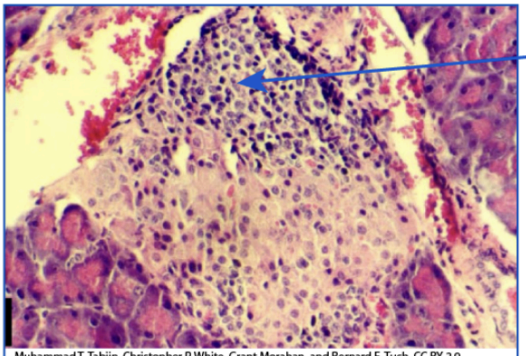

Pancreas
1
2

T1DM



Insulitis: Infiltration of insulin-producing Beta Cells by pathogenic lymphocytes

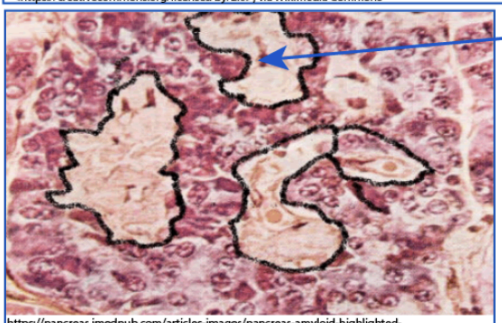
DIABETES MELLITUS



Muhammad T. Tabiin, Christopher P. White, Grant Morahan and Bernard E. Tuch, CC BY 2.0 <https://creativecommons.org/licenses/by/2.0>, via Wikimedia Commons

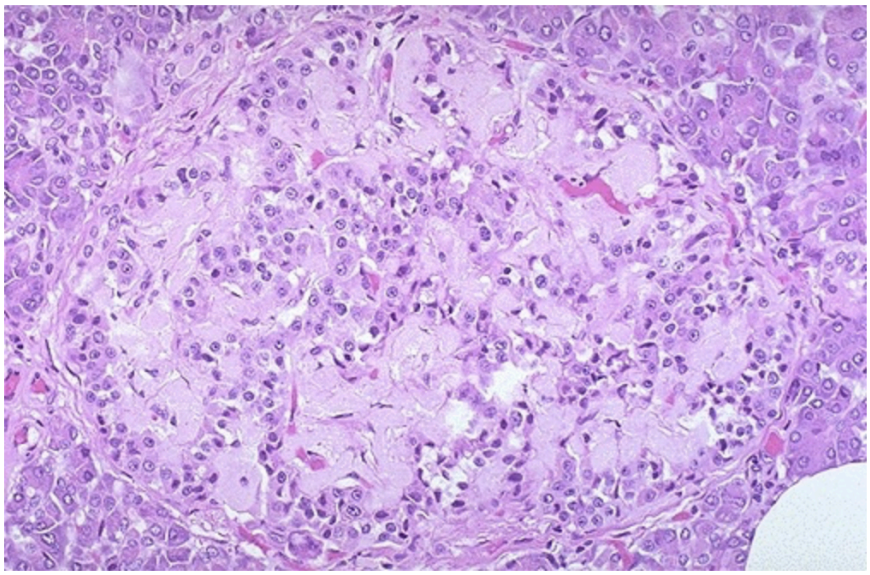
Inflammatory infiltrate

Insulitis in mouse pancreas w/ Type 1 diabetes.

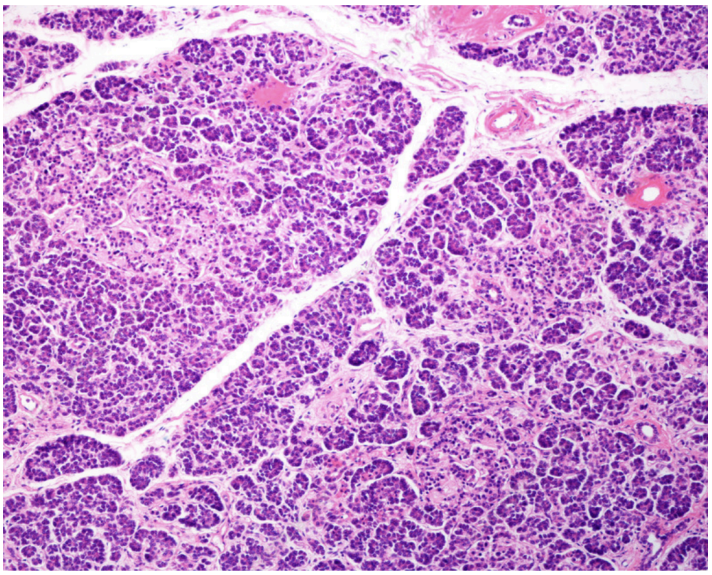


https://pancreas.imedpub.com/articles-images/pancreas-amyloid-highlighted-within-3-5-2-g001.png
Islet Amyloid, Metabolic Syndrome, and the Natural Progressive History of Type 2 Diabetes Mellitus. Melvin R Hayden

Amyloid deposits in pancreatic islet in Type 2 diabetes.



This islet of Langerhans demonstrates **pink hyalinization** (with deposition of amyloid) in many of the islet cells. This change is common in the islets of patients with type II diabetes mellitus. This amyloid contains protein derivatives of amylin, or islet amyloid polypeptide. Amylin is secreted in small amounts at the same time as insulin and slows the rate of rise of glucose in the blood.



Pancreas in type 2 diabetes mellitus, light micrograph

Vessels

- **Non-enzymatic glycation (HbA1c) and AGE accumulation** induce endothelial injury, nitric oxide depletion (impaired vasodilation), and capillary basement membrane thickening
- **Lumen narrowing** causes capillary closure and microvascular ischemia/vasa nervorum
- **Endothelial injury + dyslipidemia + insulin resistance + pro-inflammatory cytokines (IL-6, TNF-α, CRP) ↑ atherosclerosis** via leukocyte adhesion/smooth muscle proliferation → coronary artery disease/MI, stroke, and peripheral arterial disease

Kidneys

- Early glucose overload **constricts efferent arterioles (Angiotensin II)** → intraglomerular hyperfiltration
- **AGE-mediated oxidative stress thickens the glomerular basement membrane**, expands mesangium (Kimmelstiel-Wilson nodules), and detaches podocytes
- **TGF-β signaling drives glomerulosclerosis** and tubulointerstitial fibrosis → falling eGFR + progressive albuminuria
- Plasma glucose ≥180 mg/dL exceeds renal tubular reabsorption threshold → glucosuria, osmotic diuresis, and severe dehydration

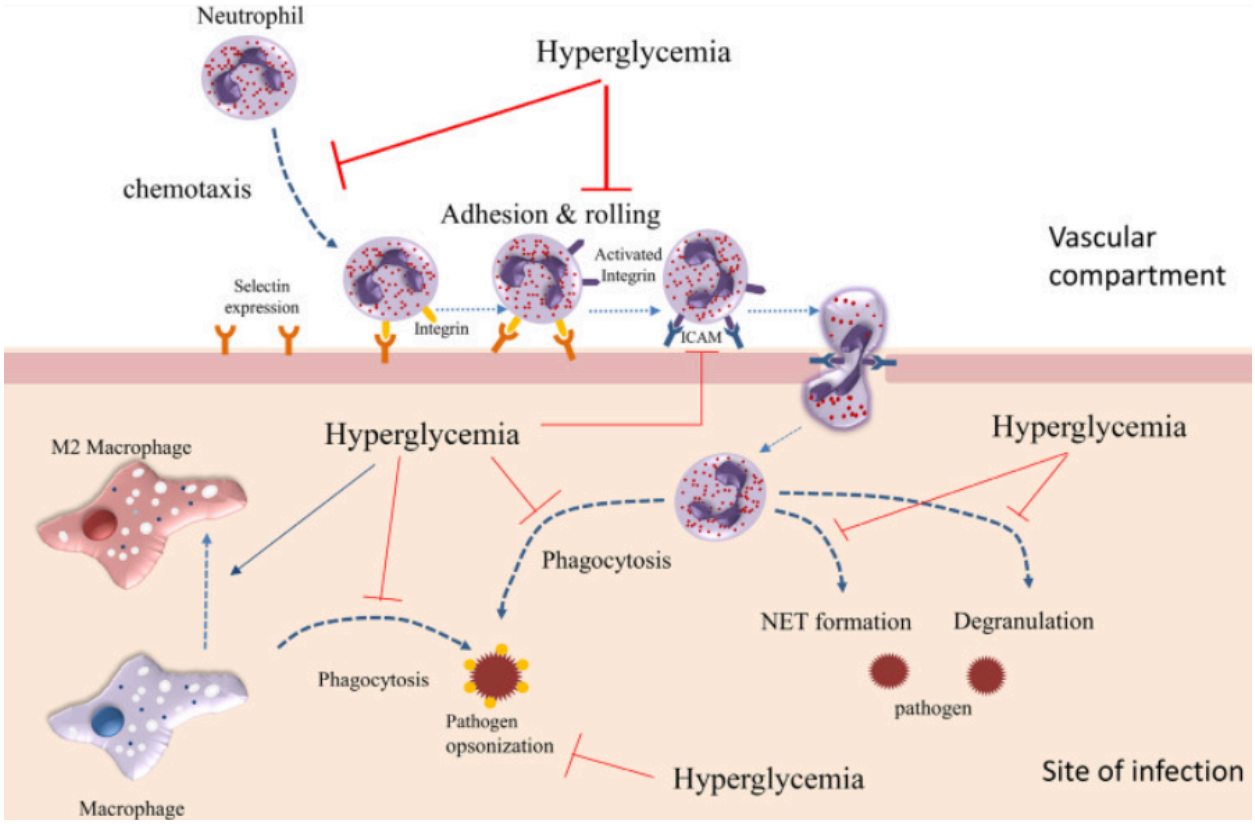
Eyes

- **Hyperglycemia, pericyte apoptosis, and basement membrane thickening** cause capillary closure and microinfarcts
    - **Non-proliferative features:** Microaneurysms, dot-and-blot hemorrhages, yellow hard lipid exudates, and cotton-wool spots (nerve fiber microinfarcts)
    - **Proliferative features:** Retinal neovascularization (fragile new vessels due to ischemia-induced VEGF upregulation), vitreous hemorrhage, and tractional retinal detachment
- **Diabetic macular edema** (retinal thickening at the macula; leading cause of vision loss in diabetic adults) is a classic clinical symptom

Nerves

- **Excess glucose shunted to sorbitol** via aldose reductase → osmotic stress, Na^+/K^+-ATPase pump failure, and NADPH depletion (oxidative stress)
- **Microvascular ischemia of vasa nervorum** starves nerve fibers → mitochondrial ATP failure and axonal/Schwann cell degeneration
- Classic clinical manifestations include **stocking-glove neuropathy** (high risk for foot ulcers and non-traumatic lower-limb amputations) **and autonomic neuropathy** (risk of resting tachycardia, gastroparesis, and neurogenic bladder)

- High ambient glucose directly impairs **neutrophil chemotaxis, phagocytosis, intracellular bacterial killing**, and reduces **natural killer (NK) cell cytotoxicity** which delays complement cascade activation (interferes with innate immunity!)
- **Non-enzymatic glycation of immunoglobulins (antibodies)** alters protein structures which impairs antigen binding and pathogen neutralization**,** which **suppresses downstream T-helper cell proliferation and cytokine production** (interferes with adaptive immunity!)
- Hyperglycemia promotes **M1 macrophage polarization** and activates **NLRP3**, elevating pro-inflammatory cytokines (**IL-1**β, **TNF-**α) which results in **persistent low-grade tissue inflammation + defective pathogen clearance** → delayed wound healing and chronic ulceration
- Systemic viral or bacterial infections (ex: flu or UTIs) serve as major physiological stress which can trigger acute **Diabetic Ketoacidosis (DKA)**
- Treatment-related infection risks (not sure if this is relevant, wanted to include just in case!):
    - Drug-induced glucosuria (**SGLT2 inhibitors)** increases glucose levels in the urinary tract, driving a **2–4× increased incidence of vulvovaginal candidiasis, UTIs**, and **Fournier's gangrene**
    - **Teplizumab (Anti-CD3 Monoclonal Antibody) d**elays T1D progression by inhibiting T-cell-mediated β-cell destruction, but carries risks of **lymphopenia and increased infection susceptibility**



1. Explain the effect of acute hyperglycemia on cognition. BCM

- Acute hyperglycemia → a short-term rise in blood glucose, over minutes to hours
    - Measurably impair thinking.
    - Effects are generally subtler and less consistent than those of hypoglycemia
    - Usually reverse once glucose returns toward normal.
    - Range from mild slowing at moderately high levels to delirium and coma in severe hyperglycemic emergencies.
    - Impair cognition mainly when it becomes severe enough to cause hyperosmolality and dehydration.
        - Progression is hyperglycemia → osmotic diuresis → water and electrolyte loss → reduced cerebral function.
- **Mild to moderate hyperglycemia.**
    - Diabetics → raising glucose to 15–17 mmol/L (roughly 270–300 mg/dL) is associated with:
        - **Slower information processing.** Reaction times and mental speed decline.
        - **Weaker working memory and attention.** Tasks that require holding and manipulating information suffer most.
        - **Mood changes.** People report more fatigue, irritability, and low mood, and these can compound the cognitive effects.
        - Lethargy and confusion
        - Altered mental status
- **Variability between people**
- **Severe hyperglycemia.**
    - Reduced level of consciousness
    - Coma in severe cases
    - Diabetic ketoacidosis (DKA) and hyperosmolar hyperglycemic state (HHS) cause confusion, drowsiness, delirium, and potentially coma.
        - mental status is often initially preserved, but severe dehydration, acidosis, or progression of the illness can also cause altered consciousness.
        - After an acute central nervous system insult, both hyperglycemia and hypoglycemia worsen neurologic outcomes
        - Target glucose of 140–180 mg/dL for nontraumatic acute CNS injury.
        - In children, DKA episodes have been linked to lasting subtle memory deficits and brain changes on imaging, so severe episodes are not always fully benign.
    - hyperglycemic hyperosmolar state (HHS) → mental status tracks serum osmolality more closely than glucose itself.
        - Difficulty concentrating or slowed thinking, Lethargy and confusion, AMS, Reduced consciousness, Coma → especially characteristic of HSS, in which glucose is often ≥600 mg/dL and dehydration is profound.
        - Patients may also develop blurred vision or weakness.
- **Acute illness.**
    - Stress hyperglycemia in hospitalized patients is associated with higher rates of delirium and worse neurological outcomes after stroke and brain injury.

    - Hard to separate the effect of glucose from the severity of the underlying illness.
- **Proposed mechanisms**
    - **Osmotic shifts.**
        - High glucose pulls water out of cells, including brain cells, which disrupts neuronal function. This is the dominant mechanism in HHS.
    - **Oxidative stress and inflammation.**
        - Excess glucose increases mitochondrial production of reactive oxygen species and activates inflammatory pathways in neurons and glia.
    - **Vascular effects.**
        - Hyperglycemia impairs endothelial function and can alter cerebral blood flow and its regulation.
    - **Altered neurotransmission.**
        - Glucose levels influence neuronal excitability and neurotransmitter systems. The hippocampus, which is central to memory, appears particularly sensitive.
    - **Metabolic and acid–base disturbances.**
        - In DKA, acidosis, ketones, and electrolyte shifts contribute independently of glucose.
- **Hypoglycemia is the bigger acute threat.** The brain depends on a continuous glucose supply, so low glucose impairs cognition faster and more severely than high glucose does.
- **The effects are reversible.**
    - In the mild-to-moderate range, function generally recovers as glucose normalizes.
    - Correction should still be gradual in severe cases, because rapid osmotic shifts carry their own risk, notably cerebral edema in pediatric DKA.
- **Acute effects differ from chronic ones.**
    - Long-term poor glycemic control is linked to a separate set of problems: accelerated cognitive decline and higher dementia risk, which work largely through vascular damage.
    - These are distinct from the acute, transient effects described above.
- **There are real-world consequences** → Even modest impairments can matter for exams, complex work tasks, or driving. This is one reason some people with diabetes notice they "think less clearly" when their glucose runs high.

1. Identify the risk factors for type 2 diabetes and summarize the recommendations regarding screening among asymptomatic individuals. CM and EBM **Sophie**

Non-Modifiable Risk Factors:

- Family history (strong genetic association, however the particular genes that cause it have not been identified)
- Race: Native, Hispanic, Black, and Asian populations
- Age >45
- History of gestational diabetes
- History of PCOS
- Prediabetes and insulin resistance

Modifiable(ish) Risk Factors

- High visceral adiposity
    - Frequently approximated by BMI but BMI is not a good way to estimate this
- Dyslipidemia
    - Especially high triglycerides and low HDL
- HTN
- Acanthosis nigricans
    - (this is associated with metabolic syndrome and insulin resistance, frequently confers risk/susceptibility to diabetes)
- Sedentary lifestyle
- Diet with lots of processed foods (sugars, fats)
- Some medications
    - Glucocorticoids
    - Atypical antipsychotics (olanzapine etc)

Screening recommendations (taken from ADA 2026 guidelines):

- Screen all adults starting no later than age **35 regardless of risk factors**
- Screen at **any age** if the individual is **overweight** (BMI>25) and has at least **1 additional risk factor**
    - *FYI adults are considered overweight if their BMI is >25, in Asian populations the threshold is >23*
- Repeat screening **every 3 years**
- ***USPSTF recommends only screening overweight adults 35-70***

Screening tests:

- HBA1c
- Fasting plasma glucose
- Oral glucose tolerance test (2 hours post 75g glucose challenge)

1. Describe the epidemiology of diabetes in the United States. CM **Sophie**

Check out this website if you have a chance! https://diabetesatlas.org/

Source: https://www.cdc.gov/diabetes/php/data-research/index.html

- In 2023, approximately 40 million people have diabetes in the US
- Among adults >18 years old, approximately 15% have diabetes
- The CDC estimates 29 million of these people are diagnosed, and 11 million are undiagnosed
- 115.2 million people in the >18 US have prediabetes
    - 31 million people aged 65 or older have prediabetes which is 52% of the population >65
- 2.1 million people have Type 1
    - 310,000 of those people are <18
- Sex:
    - Men more likely than women (15% v 12%)
- Race:
    - Highest rates in non-Hispanic black people (20.7%) followed by Hispanic people (17%) followed by Asian people (14.5%), followed by white people (11.2%)
- Education
    - Highest rates in those with less than a high school degree (23.2%), followed by high school degree (14.1%) followed by more than a high school degree (12%)

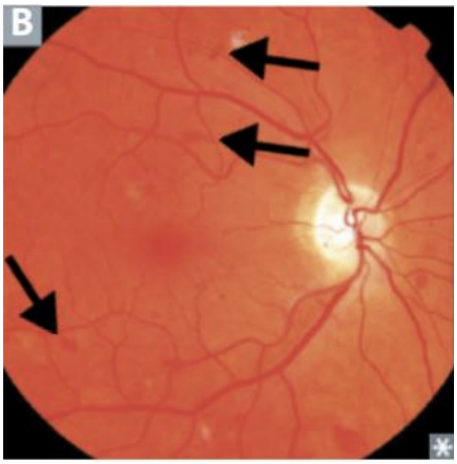
Age-Adjusted Prevalence of Diagnosed, Undiagnosed, and Total Diabetes Among Adults Aged 18 Years or Older, United States, 2021–2023

Characteristic	Diagnosed Diabetes Percentage (95% CI)	Undiagnosed Diabetes Percentage (95% CI)	Total Diabetes Percentage (95% CI)
Total	9.8 (8.5–11.3)	3.7 (2.9–4.7)	13.5 (12.0–15.2)
Sex			
Men	10.6 (9.4–12.0)	4.3 (3.0–6.0)	14.9 (13.2–16.8)
Women	9.1 (7.6–10.9)	3.3 (2.5–4.3)	12.4 (10.3–14.7)
Race and Ethnicity			
White, non-Hispanic	8.2 (7.0–9.6)	3.0 (2.2–4.1)	11.2 (9.4–13.2)
Black, non-Hispanic	15.6 (12.3–19.6)	5.2 (2.8–9.5)	20.7 (16.1–26.3)
Asian, non-Hispanic	7.9 (5.6–11.2)	6.6 (3.5–12.3)	14.5 (10.5–19.8)
Hispanic	12.9 (9.8–16.8)	4.2 (2.7–6.5)	17.1 (13.5–21.5)
Education			
Less than high school	16.9 (13.5–20.9)	6.3 (3.7–10.6)	23.2 (18.3–28.9)
High school	10.8 (8.6–13.5)	3.3 (2.4–4.7)	14.1 (11.6–17.2)
More than high school	8.3 (7.1–9.7)	3.7 (2.6–5.2)	12.0 (10.2–14.1)

Describe and interpret the common complications of diabetes seen on physical exam (ex: funduscopic exam, peripheral neuropathies, skin exam findings). CS ISHAAN

Fundoscopic exam:

Non-proliferative (most common): microaneurysms, hemorrhage, cotton wool spots, hard exudates. Vision loss mainly due to macular edema.



Arrows are pointing at hemorrhage

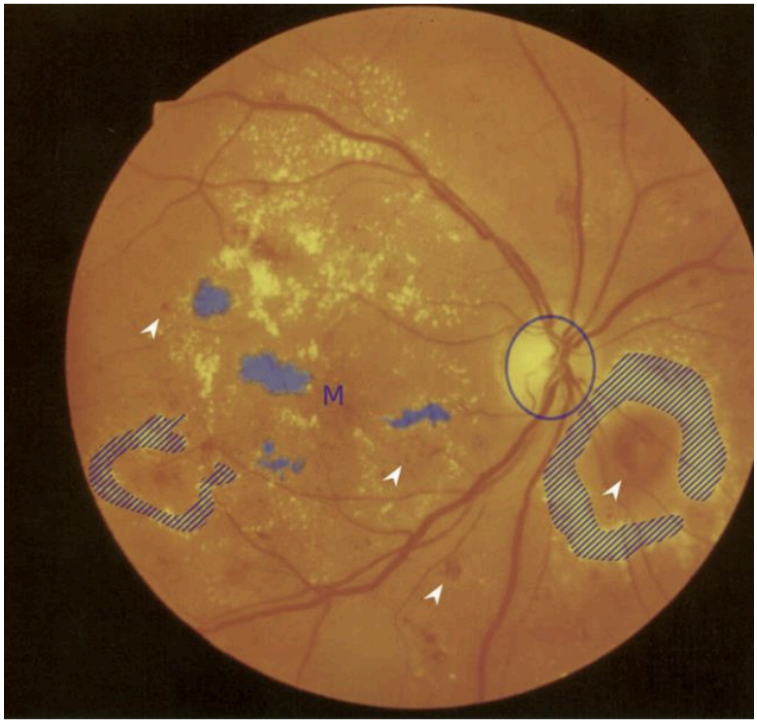
You can also see cotton wool spots (the white blurs, one is very close to the top arrowhead)

Hard exudates appear as sharply demarcated, yellow lesions (examples marked by blue overlay).

When arranged in a circle, exudates are called circinate rings (hatched overlay).

Microaneurysms and retinal bleedings appear as small red dots and larger red blotches (examples marked by arrowheads).

Neovascularizations are absent.
Blue circle: Optic disc
M: Macula



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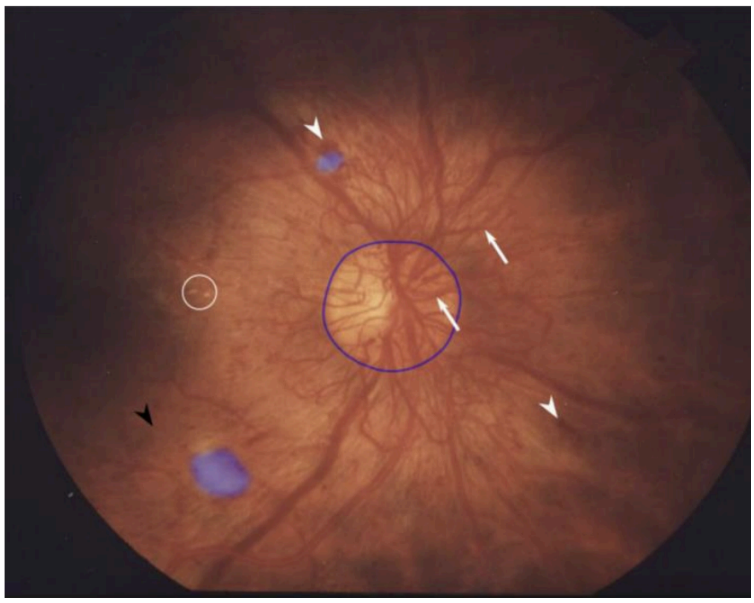
Proliferative: retinal neovascularization due to chronic hypoxia, abnormal new vessels may lead to vitreous hemorrhage + traction retinal detachment

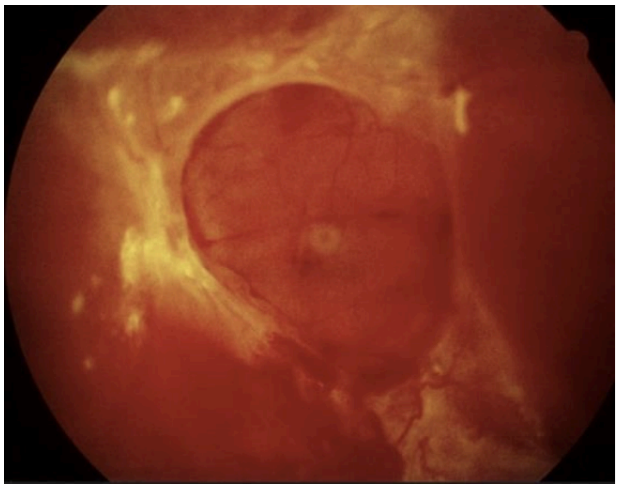
There is massive neovascularization (examples indicated by arrows) of the optic disk (NVD), cotton wool spots (overlay), hard exudates (circle), intraretinal hemorrhages (examples indicated by white arrowheads), and a microaneurysm (example indicated by black arrowhead).

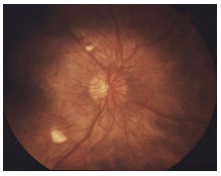
The diagnosis of proliferative disease requires the presence of neovascularization.

Center outline = optic disc



Traction retinal detachment: → fibrovascular proliferation (light orange to left of big red circle) → tractional retinal detachment + retinal hemorrhages seen in dark red spots at bottom

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Peripheral neuropathies:

Sensory symptoms:

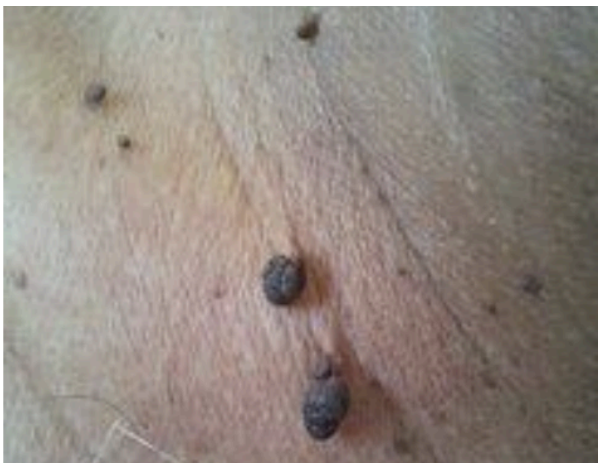
- Distal symmetrical sensory loss: stocking glove sensory loss pattern with proximal progression
- Dysesthesia (e.g., burning feet syndrome with symptoms that worsen at night)
- Motor weakness

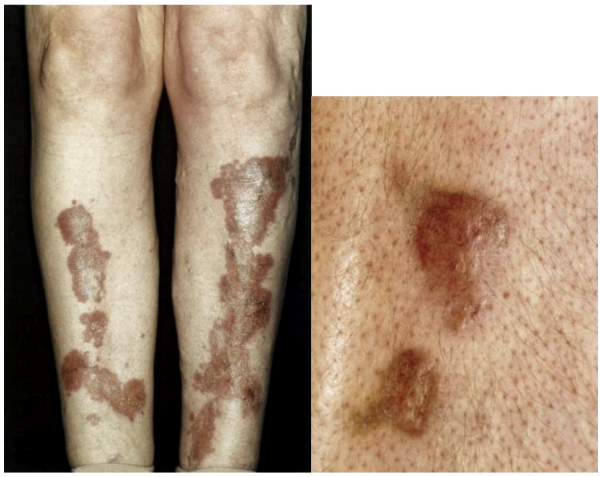
Neurological examination may show loss or reduction of:

- Vibration sense (using a tuning fork)
- Monofilament test: can be used to detect diminished sensory perception (peripheral neuropathy) on diabetic foot exam.

- The monofilament is a single-fiber nylon strand attached to a plastic base. It's applied to the patient's skin to generate a reproducible buckling stress → the examiner asks the patient to close their eyes and to say when they feel the sensation → examiner applies the monofilament with gentle pressure on at least 4 points on the sole of the foot: the 1st, 3rd, and 5th metatarsal heads, and the distal hallux

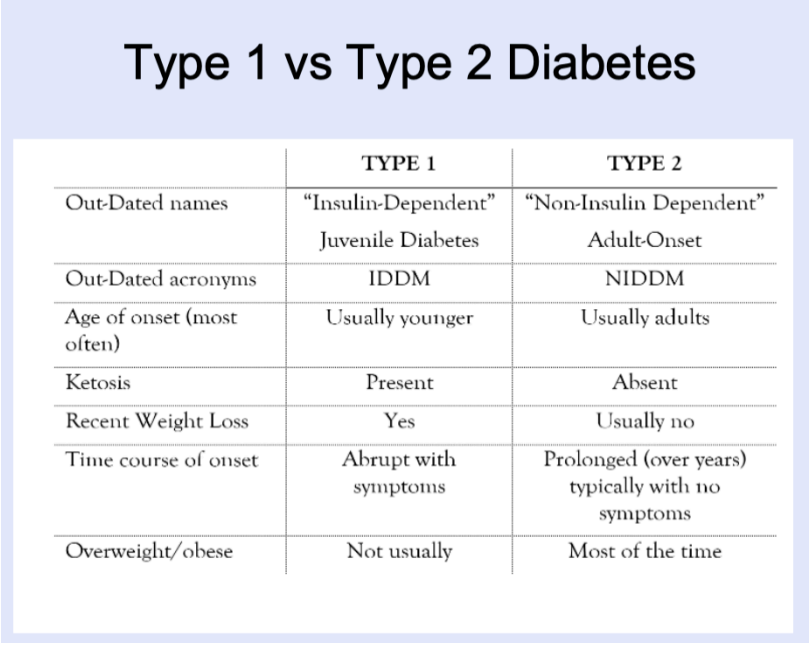
Skin exam:

Acanthosis nigricans: hyperpigmented skin and hyperkeratosis are visible in this patient's left axilla (may also be seen in groin or neck). An erythematous patch is visible in between the hyperpigmented areas Seen in T2DM due to insulin resistance	
Acrochordons (skin tags): darkcolored, pedunculated papules Often found on back of neck or armpits	

Necrobiosis lipoidica: chronic complication of diabetes; inflammatory granulomatous disorder of the skin resulting in collagen degeneration and lipid accumulation in the surface of the skin	
Cellulitis: skin + soft tissue infection (often seen on feet)	
Diabetic foot ulcers: Skin breakdown with or without surrounding tissue necrosis Neuropathic ulcers occur at sites of repetitive stress/trauma, such as: - head of the metatarsal bones or the heel.	

1. Compare and contrast the clinical presentation for type 1 and type 2 diabetes. CM- **Tomiwa**

Feature	Type 1 DM	Type 2 DM
Onset	Usually rapid	Usually gradual/insidious
Typical patient	Often younger, frequently thin	Often overweight/obese, especially central obesity
Symptoms	Polyuria, polydipsia, weight loss, fatigue, visual disturbances	Polyuria, polydipsia, fatigue; may be asymptomatic initially
Hyperglycemic crisis	DKA is typical	HHS is typical
DKA findings	Nausea/vomiting, abdominal pain, dehydration, Kussmaul respirations, fruity breath	Can occur, but less typical
HHS findings	Less typical	Profound dehydration, altered mental status, severe hyperglycemia



Type 1 vs Type 2 Diabetes

	TYPE 1	TYPE 2
Insulin Resistance	No	Yes
Islet Cell Antibodies & GAD-65 Antibodies	Yes	No
Insulin Secretion Defect	Yes	Yes
HLA-Associated Antigens	Yes	No
Associated with other Auto-immune Diseases	Yes	No
Insulin Rx Necessary	Absolutely	Sometimes

3. Compare and contrast the clinical presentations of the diabetic emergencies (DKA vs HHS). CM - **Ayla P**

	DKA	HHS
Associated with	Type 1 diabetes (insulin deficiency)	Type 2 diabetes (insulin resistance)
Typical patient	- Child/young adult - Thin body habitus	- Adult - Often overweight/obese - Other signs of metabolic syndrome may be present
Labs	- Elevated plasma glucose (>250) - Elevated pOsm (>290) - Acidosis (pH < 7.30) - [HCO3-] > 18 - Elevated beta-hydroxybutyrate (>3 mmol/L) - Ketonuria	- Very high plasma glucose (>600) - High pOsm (>320) - pH >7.30 - Negative/trace ketones in plasma and urine
Symptoms	- Polydipsia, polyuria, weakness - Nausea/vomiting - Abdominal pain - Kussmaul respirations - Fruity (acetone) odor of breath	- Polydipsia, polyuria, weakness - Profound dehydration - Altered mental status

Recognize values for plasma glucose, bicarbonate, ketone bodies, osmolality, and blood gases that are typical of HHS. BCM - **Spencer**

- Plasma glucose: >600 mg/dL
- Effective serum osmolality: >320 mOsm/kg
    - 2 × Na^+ + glucose (mmol/L); BUN excluded (urea crosses membranes freely)
- Bicarbonate: >15-18 mmol/L (US ≥18; UK ≥15), so no significant acidosis
- Ketones (β-hydroxybutyrate): <3.0 mmol/L; urine/serum ketones negative or low-positive
- Blood gas (pH): >7.30 (normal or near-normal)
- Anion gap: normal (<12 mmol/L)

Key concepts

- Residual insulin suppresses ketogenesis but is insufficient for glucose uptake → severe hyperglycemia → osmotic diuresis → dehydration and hyperosmolality, without ketoacidosis.
- Mixed DKA-HHS overlap occurs (up to ~1/3 of cases; ~20-30% of HHS patients have some increased-AG acidosis), so mild ketonemia does not exclude HHS.

1. List and recognize the current ADA criteria for the diagnosis of diabetes.

- The American Diabetes Association (ADA) recommends that diabetes may be diagnosed using A1C or plasma glucose criteria, with any of four tests being acceptable for diagnosis.
- KNOW THIS: Meeting any **one of the following** thresholds satisfies the diagnostic criteria:
    - **Fasting plasma glucose (FPG) ≥126 mg/dL (≥7.0 mmol/L)**
        - fasting defined as no caloric intake for at least 8 hours.
    - **2-hour plasma glucose (2-h PG) ≥200 mg/dL (≥11.1 mmol/L)** during a 75-g oral glucose tolerance test (OGTT).
    - **A1C ≥6.5% (≥48 mmol/mol)**.
    - **Random plasma glucose ≥200 mg/dL (≥11.1 mmol/L)**
        - Pt with classic symptoms of hyperglycemia (polyuria, polydipsia, unexplained weight loss) or a hyperglycemic crisis (DKA OR HSS).
- FPG, 2-h PG, and A1C are all **appropriate for both screening and diagnosis**, but reflect different aspects of glucose metabolism and have incomplete concordance
- compared with FPG and A1C cut points, the **2-h PG value identifies more people** with prediabetes and diabetes.

Confirming the diagnosis: Extra but helpful

- In the **absence of a clear clinical diagnosis** (e.g., classic symptoms or hyperglycemic crisis with random glucose ≥200 mg/dL), **confirmatory testing is required**.
    - Accomplished either by two abnormal results from the same sample, or two abnormal results from samples collected at two different time points.
- Two **different tests** (e.g., A1C and FPG) both above the diagnostic threshold (whether drawn at the same time or at two different times) confirm the diagnosis.
- If results are **discordant**, repeat the test whose result is above the diagnostic cut point; the diagnosis is based on the confirmed test.
    - Ex., two A1C results ≥6.5% confirm diabetes even if FPG is <126 mg/dL.
- Results **close to the threshold** warrant patient education on hyperglycemia symptoms and repeat testing in 3–6 months. Persistent, substantial discordance between glucose and A1C should prompt evaluation for analytic interference.

The following table summarizes the four diagnostic criteria: Includes prediabetes

Test	Diabetes threshold	Prediabetes range
Fasting plasma glucose (FPG)	≥126 mg/dL (≥7.0 mmol/L)	100–125 mg/dL (IFG)
2-h plasma glucose (75-g OGTT)	≥200 mg/dL (≥11.1 mmol/L)	140–199 mg/dL (IGT)

A1C	≥6.5% (≥48 mmol/mol)	5.7–6.4%
Random plasma glucose	≥200 mg/dL (≥11.1 mmol/L) + classic symptoms/crisis	N/A

A few additional clinical notes: Probably important

- **Prediabetes**
  - FPG **100–125** mg/dL (impaired fasting glucose), 2-h PG **140–199** mg/dL (impaired glucose tolerance), or A1C **5.7–6.4**%.
- **A1C limitations**: A1C is not reliable in conditions that alter the relationship between A1C and glucose (e.g., hemoglobinopathies, pregnancy, recent transfusion, conditions with altered red cell turnover) or in situations of rapid glucose change such as early type 1 diabetes.
- **Note on type 1 vs. type 2**: These glucose/A1C criteria establish the diagnosis of diabetes but do not distinguish type.
  - Autoantibody testing (e.g., GAD65, IA-2, ZnT8, insulin) helps differentiate type 1 from type 2.

1. Define the condition of prediabetes and describe the risk of progression to diabetes. CM **Sophie**

Prediabetes is defined by having ONE of these three things:

- **a HbA1c** between 5.7 and 6.4
- **OGTT** (oral glucose tolerance test) between 140-199mg/dL
- **fasting glucose** between 100-125mg/dL

52% of the population >65yo is prediabetic

5-10% of people with prediabetes every year progress to T2DM

- Approximately 40% of people progress to diabetes over 25 years (35-50% progress over 5-10 years)(not great consensus on this)

Risk factors for progression are the same as T2DM Risk Factors in Epi 1:

- Higher HbA1c, fasting glucose, OGTT
    - Having positives in multiple of these tests rather than just one
- Higher BMI
- Inactivity
- Fam history of T2DM
- Gestational diabetes
- PCOS
- Old age
- HTN
- Dyslipidemia
- Insulin resistance/metabolic syndrome

Prediabetes can reverse!!!

- Focus on modifiable risk factors: weight management, dyslipidemia management, sedentary lifestyle, diet, HTN management

1. Describe the major effect of metformin on carbohydrate metabolism.

- Major effect of metformin = **suppression of hepatic glucose production** → inhibiting **gluconeogenesis** in the liver.
    - lowers fasting blood glucose, since the liver is the main source of glucose between meals.
- **How it does this**
    - **Mitochondrial effects:**
        - Metformin accumulates in hepatocytes and mildly inhibits mitochondrial respiratory chain complex I.
        - Also inhibit mitochondrial glycerophosphate dehydrogenase.
            - Changes the cell's energy and redox state, raising the AMP:ATP ratio and the cytosolic NADH:NAD^+ ratio.
    - **Less substrate flowing into glucose:**
        - The altered redox state limits the conversion of lactate and glycerol into glucose.
    - **AMP-related signaling:**
        - Higher AMP inhibits fructose-1,6-bisphosphatase, a key gluconeogenic enzyme.
        - It also blunts glucagon signaling by inhibiting adenylate cyclase, which reduces cAMP and PKA activity.
        - It activates AMP-activated protein kinase (AMPK), which lowers expression of gluconeogenic genes and reduces hepatic lipogenesis.
- **Secondary effects**
    - Increases insulin sensitivity, with modestly greater glucose uptake in skeletal muscle.
    - Increases glucose use by the intestine, partly through anaerobic glycolysis. (increase glycolysis)
        - Why gut glucose uptake rises on PET imaging.
    - Modestly raises GLP-1 and alters the gut microbiome.
    - Reduces intestinal glucose absorption
- **Clinical consequences**
    - Metformin does **not stimulate insulin secretion**.
        - Rarely causes hypoglycemia when used alone, and it does not cause weight gain.
    - Because it reduces the liver's conversion of lactate to glucose, lactate can build up.
        - In patients who have other risk factors, such as significant kidney impairment, hypoxia, or sepsis, this can lead to **lactic acidosis**. This complication is rare but serious.

Compare and contrast the mechanisms of action, relative efficacies, adverse effects, and contraindications for drug classes used to treat type 2 diabetes. PHARM **Ishaan**

Drug Class / Agent	MOA	Use
Metformin *(Biguanide is the drug class but metformin is the main/only drug we need to know)* ***MET****formin = Met**abolic (latic) acidosis*	- Metformin **inhibits hepatic gluconeogenesis** - Increases glycolysis in cell tissue - **Increases peripheral glucose uptake** (increased insulin sensitivity)	- 1st-line therapy for T2DM - Weight neutral / weight loss
SGLT2 Inhibitors -flozin *Glucose floz in through the urine*	Inhibits SGLT2 in the PCT of the kidney → decreases renal glucose reabsorption → increases urinary glucose excretion	- Moderate weight loss - Reduces CHF hospitalizations, ASCVD, and CKD progression
Sulfonylureas -izide *Act on pancreatic B cells!*	Closes ATP-sensitive K+ channels on pancreatic beta cells → cell depolarization and Ca2+ influx → stimulates insulin release	- High initial HbA1c reduction - Weight gain
Meglitinides -glinide *Act on pancreatic B cells!*	Closes ATP-sensitive K+ channels on beta cells (same as sulfonylureas), but with a rapid onset and shorter duration of action (taken with meals)	- Postprandial glucose control - Expensive ($$$)
Thiazolidinediones (TZDs) -glitazone *bring glita to the PPARty*	PPAR-γ receptor agonist (on muscle, adipose, liver) → modulates gene expression → increase insulin sensitivity *Only drug to go into cell nucleus and change gene expression*	- Weight gain - Increases HDL

DPP-4 Inhibitors -liptin *-liptin → Lipton tea → tea makes you pee → causes urinary infxns → DPP*	Inhibits DPP-4 enzyme → prevents degradation of endogenous incretins (GLP-1 and GIP) → increases postprandial insulin release	
GLP-1 Receptor Agonists -tide *GULP -tide pods*	Agonist for GLP-1 (incretin) receptors → increases glucose dependent insulin secretion + suppresses postprandial glucagon + delays gastric emptying + promotes central satiety	- Significant weight loss - Reduces ASCVD events - 2nd-line for CKD risk reduction
Incretins	Gut peptides released postprandially in response to oral glucose	Endogenous foundation for GLP-1 RAs, DPP-4 inhibitors, and Dual GLP-1/GIP agonists.
Dual GLP-1RA & GIPRA (Twincretins)	Dual agonist for both GLP-1 and GIP receptors → synergistic stimulation of glucose-dependent insulin release, glucagon suppression, and satiety	- Highest HbA1c reduction - Greatest weight loss - Expensive ($$$)

3. Explain how diabetes prevention and management contribute to healthcare value. HSS (attos://www.cdc.gov/nccdphp/priorities/diabetes- interventions.htmi#fedc generic section 3-strategies-that-work) - **Ayla P**

Treatment of diabetes and diabetes-related complications is one of the US healthcare system's greatest expenditures, costing **$306 billion in total and an additional $106 billion in lost productivity**. Diabetes cases are expected to increase in future years, necessitating prevention and management strategies.

Diabetes prevention and management significantly reduce healthcare costs associated with acute hospitalizations and long-term complications, which benefits individual patients and the healthcare system as a whole:

- Individuals with prediabetes enrolled in the National Diabetes Prevention Program save an average of **$4552 in direct medical costs** in the two year period following enrollment.
- For every case of type 2 diabetes prevented by the NDPP, the **US healthcare system saves $160,000.**

Diabetes prevention/management strategies are a **good value in terms of quality-adjusted life years (QALYs):**

- Intensive lifestyle modification for high-risk patients: $12,500 per QALY
- Self-monitoring of blood glucose 3 times per day compared to once per day: $7,300 per QALY
- Eye screening every 1-2 years: $8,763 per QALY
- Annual CKD screening: $12,500 per QALY

(In general, interventions under $50,000 per QALY are considered high-value)

Diabetes Self-Management Education and Support (DMSES) is a highly evidence-based strategy to reduce complications in diabetic patients. In addition to better outcomes and fewer complications, **DMSES has many benefits for value of care**:

- Support of patients, providers, and community members through education and awareness
- Increased quality and decreased cost of health care services
- Setting standards and goals for health care reform and helping providers achieve these goals

4. Describe lifestyle strategies and their benefits in the diabetic and prediabetic patient.
WELL (see additional resources) - **Ayla P**

Lifestyle strategies for prediabetic patients are focused on preventing progression to diabetes. Strategies include:

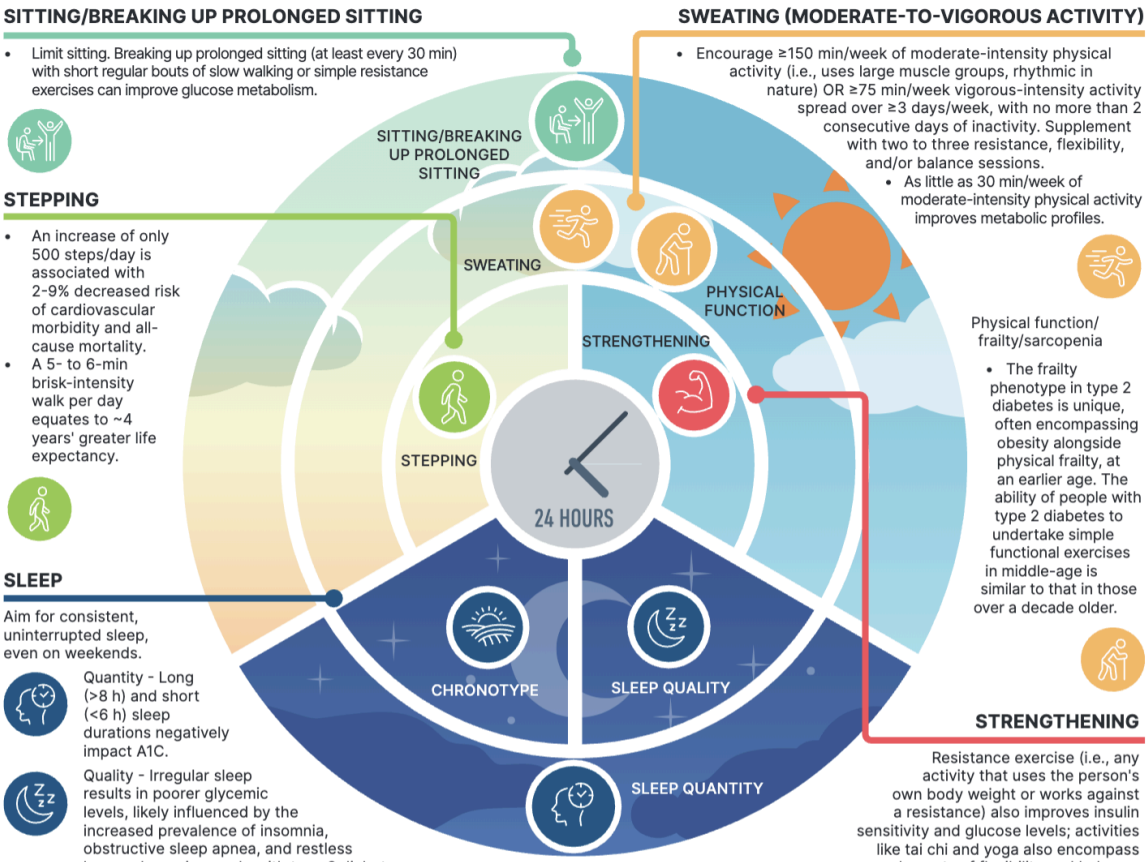
- Participation in programs like NDPP
- Weight management (5-7% reduction in body weight is recommended for high-risk patients)
- Dietary adjustments (whole-foods focused)
- Regular physical activity
- Improved sleep hygiene
- Avoidance of tobacco

Lifestyle strategies for diabetic patients are similar, and can be grouped into 6 main categories:

- Diabetes self-management education and support (DSMES)
- Medical nutrition therapy: working with a registered dietitian to establish a sustainable, realistic, flexible diet plan consisting primarily of plant-based whole foods
- Counseling on routine physical activity: cardio, strength training, overall daily steps, and breaking up prolonged periods of sitting are all associated with better outcomes
- Smoking/vaping cessation support, avoidance of cannabis
- Counseling on health behaviors/sleep quality: focusing on quality, quantity, and consistency
- Psychosocial support from behavioral health professionals: regular mental health screenings and prompt treatment

For more detail:

Importance of 24-Hour Physical Behaviors for Type 2 Diabetes



SITTING/BREAKING UP PROLONGED SITTING

- Limit sitting. Breaking up prolonged sitting (at least every 30 min) with short regular bouts of slow walking or simple resistance exercises can improve glucose metabolism.

STEPPING

- An increase of only 500 steps/day is associated with 2-9% decreased risk of cardiovascular morbidity and all-cause mortality.
- A 5- to 6-min brisk-intensity walk per day equates to ~4 years' greater life expectancy.

SLEEP

Aim for consistent, uninterrupted sleep, even on weekends.

Quantity - Long (>8 h) and short (<6 h) sleep durations negatively impact A1C.

Quality - Irregular sleep results in poorer glycemic levels, likely influenced by the increased prevalence of insomnia, obstructive sleep apnea, and restless leg syndrome in people with type 2 diabetes.

Chronotype - Evening chronotypes (i.e., night owl: go to bed late and get up late) may be more susceptible to inactivity and poorer glycemic levels than morning chronotypes (i.e., early bird: go to bed early and get up early).

SWEATING (MODERATE-TO-VIGOROUS ACTIVITY)

- Encourage ≥150 min/week of moderate-intensity physical activity (i.e., uses large muscle groups, rhythmic in nature) OR ≥75 min/week vigorous-intensity activity spread over ≥3 days/week, with no more than 2 consecutive days of inactivity. Supplement with two to three resistance, flexibility, and/or balance sessions.
  - As little as 30 min/week of moderate-intensity physical activity improves metabolic profiles.

Physical function/ frailty/sarcopenia

- The frailty phenotype in type 2 diabetes is unique, often encompassing obesity alongside physical frailty, at an earlier age. The ability of people with type 2 diabetes to undertake simple functional exercises in middle-age is similar to that in those over a decade older.

STRENGTHENING

Resistance exercise (i.e., any activity that uses the person's own body weight or works against a resistance) also improves insulin sensitivity and glucose levels; activities like tai chi and yoga also encompass elements of flexibility and balance.

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1. Describe the major acute and chronic diabetic complications that are the result of poor glucose control. CM

Acute	DKA	HHS
Who	T1DM	T2DM, elderly
Why	No insulin → lipolysis → ketones	Some insulin → no ketones; osmotic diuresis → dehydration
Labs	Glucose >250, pH <7.3, ketones +	Glucose >600, osm >320, no ketones
Classic	Kussmaul breathing, fruity breath, abd pain	Confusion/coma, severe dehydration

Chronic	Mechanism	Key findings
Retinopathy	Microvascular damage → ischemia → VEGF	Microaneurysms, hemorrhages → neovascularization
Nephropathy	GBM thickening, mesangial expansion	Kimmelstiel-Wilson nodules, albuminuria
Neuropathy	Sorbitol buildup in nerves	Stocking-glove sensory loss, gastroparesis
Macrovascular	Accelerated atherosclerosis	MI (#1 cause of death), stroke, PAD
Diabetic foot	Neuropathy + PAD	Painless ulcers → amputation